Solved selected problems of Modern Physics for Scientists and Engineers - John Taylor

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Chapter 8 - The Three-Dimensional Schrödinger Equation

Solution. 8.4 For a free particle subject to no forces, equation (8.2) becomes

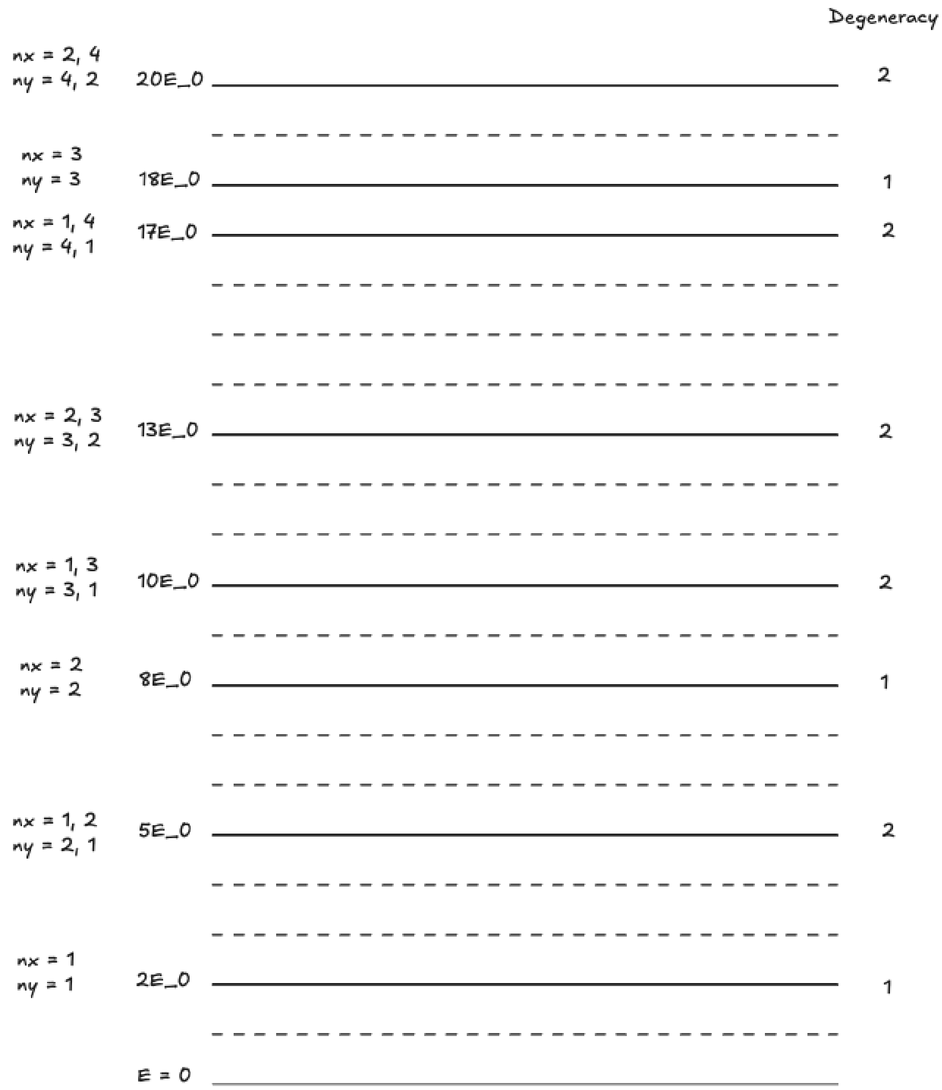
$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} = -\frac{2M}{\hbar^2}E\psi$$

We want to show that $\psi(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$ is a solution for any fixed vector $\boldsymbol{k}$ satisfying $E = \hbar^2\boldsymbol{k}^2/2M$ hence we replace ψ and E and we solve the equation

$$\begin{aligned}
-k_x^2 e^{i\boldsymbol{k}\cdot\boldsymbol{r}} - k_y^2 e^{i\boldsymbol{k}\cdot\boldsymbol{r}} - k_z^2 e^{i\boldsymbol{k}\cdot\boldsymbol{r}} &= -\frac{2M}{\hbar^2}\frac{\hbar^2\boldsymbol{k}^2}{2M}e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \\
-e^{i\boldsymbol{k}\cdot\boldsymbol{r}}(k_x^2 + k_y^2 + k_z^2) &= -\boldsymbol{k}^2 e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \\
k_x^2 + k_y^2 + k_z^2 &= \boldsymbol{k}^2
\end{aligned}$$

Where k_x, k_y and k_z are the components of the vector $\boldsymbol{k}$. We see that the equality holds, so $\psi(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$ is a solution to the three-dimensional Schrödinger equation.
The vector $\boldsymbol{k}$ can be viewed as the angular frequency vector, where each component represents the angular frequency of the wave equation in a particular direction. □

Solution. 8.8


Degeneracy
nx = 2, 4
ny = 4, 2
20E_0
2
nx = 3
ny = 3
18E_0
1
nx = 1, 4
ny = 4, 1
17E_0
2
nx = 2, 3
ny = 3, 2
13E_0
2
nx = 1, 3
ny = 3, 1
10E_0
2
nx = 2
ny = 2
8E_0
1
nx = 1, 2
ny = 2, 1
5E_0
2
nx = 1
ny = 1
2E_0
1
E = 0

□

Solution. 8.9 Consider a particle of mass M in a two-dimensional, rigid rectangular box with sides a and b. Following the same steps as in Section 8.3 from equation (8.3) to equation (8.15) everything in between applies to this case. We arrive to the following differential equations

$$X''(x) = -k_x^2 X(x) \qquad Y''(y) = -k_y^2 Y(y)$$

For which we know the solutions are

$$X(x) = B \sin k_x x \qquad Y(y) = C \sin k_y y$$

And to satisfy the boundary conditions $X(x) = 0$ at $x = 0$ or a and $Y(y) = 0$ at $y = 0$ or b we need that

$$k_x = \frac{n_x \pi}{a} \qquad k_y = \frac{n_y \pi}{b}$$

Now combining the solutions we get the complete wave function solution

$$\psi(x, y) = A \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right)$$

Where we renamed the constant BC as A.
On the other hand, from equation (8.10) we know that

$$\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} = -\frac{2ME}{\hbar^2}$$

But $X''(x)/X(x) = -k_x^2$ and $Y''(y)/Y(y) = -k_y^2$ then

$$\begin{aligned} -\frac{2ME}{\hbar^2} &= -\frac{n_x^2 \pi^2}{a^2} - \frac{n_y^2 \pi^2}{b^2} \\ E_{n_x, n_y} &= \frac{\pi^2 \hbar^2}{2M} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} \right) \end{aligned}$$

Which gives us the allowed energies. $\square$

Solution. 8.10 Let

$$E_0 = \frac{\hbar^2\pi^2}{2Ma^2}$$

Then equation (8.102) for the case $b = a/2$ becomes

$$E_{n_x,n_y} = E_0(n_x^2 + 4n_y^2)$$

Then the six lowest energy levels, with their quantum numbers and degeneracies are

E_{n_x,n_y}	Degeneracy	n_x, n_y
$5E_0$	1	$\begin{cases} n_x = 1 \\ n_y = 1 \end{cases}$
$8E_0$	1	$\begin{cases} n_x = 2 \\ n_y = 1 \end{cases}$
$13E_0$	1	$\begin{cases} n_x = 3 \\ n_y = 1 \end{cases}$
$17E_0$	1	$\begin{cases} n_x = 1 \\ n_y = 2 \end{cases}$
$20E_0$	2	$\begin{cases} n_x = 2, 4 \\ n_y = 2, 1 \end{cases}$
$25E_0$	1	$\begin{cases} n_x = 3 \\ n_y = 2 \end{cases}$

$\square$

Solution. 8.11 Let $a = 1.1b$ then the equation (8.102) for this case becomes

$$E_{n_x,n_y} = E_0\left(\frac{n_x^2}{(1.1)^2} + n_y^2\right)$$

Then the six lowest energy levels, with their quantum numbers and degeneracies are

E_{n_x,n_y}	Degeneracy	n_x, n_y
$1.82E_0$	1	$\begin{cases} n_x = 1 \\ n_y = 1 \end{cases}$
$4.3E_0$	1	$\begin{cases} n_x = 2 \\ n_y = 1 \end{cases}$
$4.82E_0$	1	$\begin{cases} n_x = 1 \\ n_y = 2 \end{cases}$
$7.3E_0$	1	$\begin{cases} n_x = 2 \\ n_y = 2 \end{cases}$
$8.43E_0$	1	$\begin{cases} n_x = 3 \\ n_y = 1 \end{cases}$
$9.82E_0$	1	$\begin{cases} n_x = 1 \\ n_y = 3 \end{cases}$

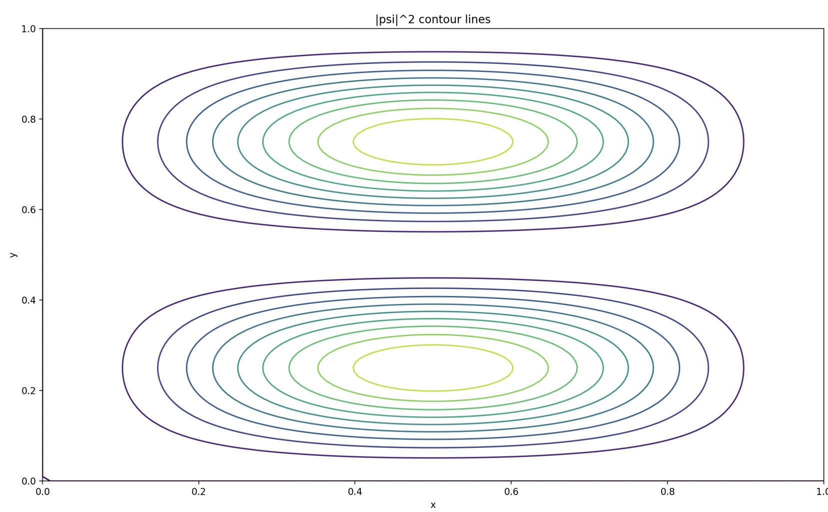
We see that the degeneracies $n_x = 1, 2, n_y = 2, 1$ and $n_x = 3, 1, n_y = 1, 3$ disappear. □

Solution. 8.12

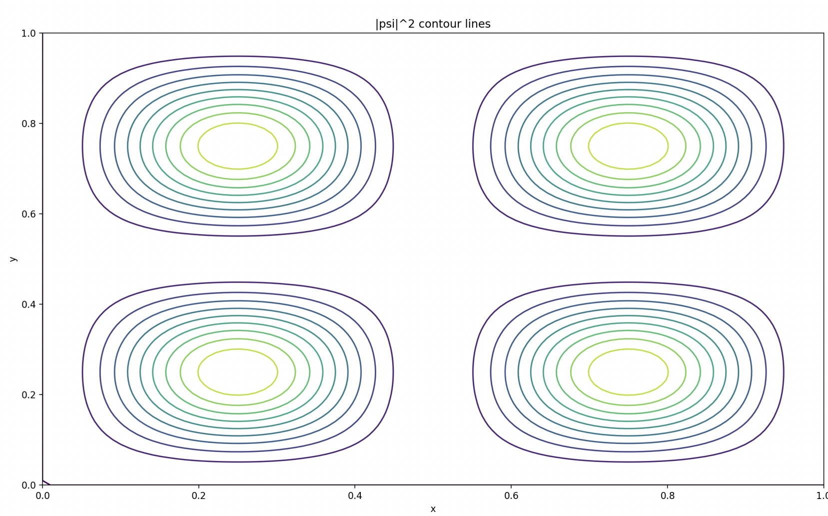
(a) Let $n_x = 1$ and $n_y = 2$ then

$$|\psi(x, y)|^2 = A^2 \sin^2 \frac{\pi x}{a} \sin^2 \frac{2\pi y}{a}$$

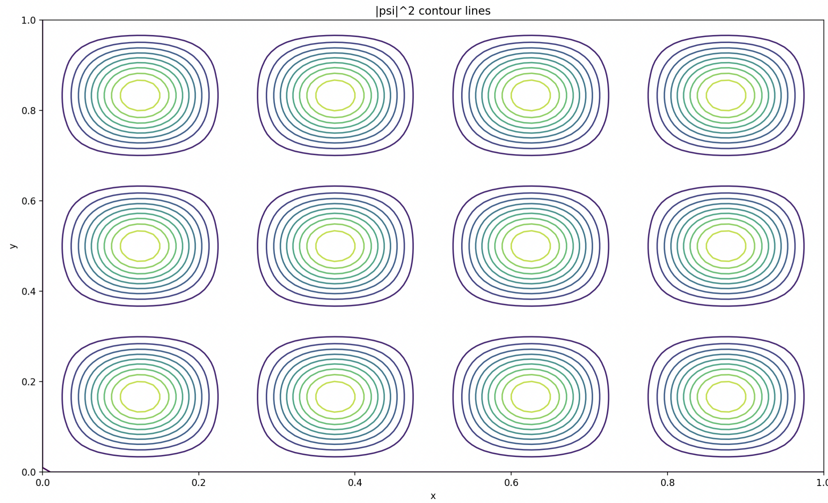
Then the maximum of $|\psi|^2$ happen at $x = a/2$ and $y = a/4, 3a/4$ so we have 2 points where the particle is most likely to be found. Below we show the contour map for this case



(b) Let $n_x = 2$ and $n_y = 2$ then the contour map looks like



(c) Let $n_x = 4$ and $n_y = 3$ then the contour map looks like



$\square$

Solution. 8.22

(a) Let $\psi = R(r)\Theta(\theta)\Phi(\phi)$, then, replacing in the Schrödinger equation for spherical coordinates we get that

$$\frac{\Theta\Phi}{r}\frac{\partial^2}{\partial r^2}(rR) + \frac{R\Phi}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right) + \frac{R\Theta}{r^2\sin^2\theta}\frac{\partial^2\Phi}{\partial\phi^2} = \\ = \frac{2M}{\hbar^2}[U(r) - E]R\Theta\Phi$$

Multiplying the equation by $r^2\sin^2\theta/(R\Theta\Phi)$ we have that

$$\frac{r\sin^2\theta}{R}\frac{\partial^2}{\partial r^2}(rR) + \frac{\sin\theta}{\Theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right) + \frac{1}{\Phi}\frac{\partial^2\Phi}{\partial\phi^2} = \\ = \frac{2M}{\hbar^2}[U(r) - E]r^2\sin^2\theta$$

Rearranging

$$\frac{1}{\Phi}\frac{\partial^2\Phi}{\partial\phi^2} = \frac{2M}{\hbar^2}[U(r) - E]r^2\sin^2\theta - \frac{r\sin^2\theta}{R}\frac{\partial^2}{\partial r^2}(rR) - \frac{\sin\theta}{\Theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right)$$

So we got

$$\frac{\Phi''(\phi)}{\Phi(\phi)} = \text{function of } r \text{ and } \theta$$

Since the RHS is independent of ϕ and the LHS is independent of r and θ they must be equal to a constant, otherwise a change in the LHS would imply a change in the RHS to maintain the equality, which doesn't make sense since they are independent.
Naming the constant $-m^2$ gives us

$$\Phi''(\phi) = -m^2\Phi(\phi)$$

(b) On the other hand, this implies that

$$\frac{2M}{\hbar^2}[U(r) - E]r^2\sin^2\theta - \frac{r\sin^2\theta}{R}\frac{\partial^2}{\partial r^2}(rR) - \frac{\sin\theta}{\Theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right) = -m^2$$

Multiplying the equation by $1/\sin^2\theta$ gives us

$$\frac{2M}{\hbar^2}[U(r) - E]r^2 - \frac{r}{R}\frac{\partial^2}{\partial r^2}(rR) - \frac{1}{\Theta\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right) = -\frac{m^2}{\sin^2\theta}$$

And rearranging it we get

$$\frac{1}{\Theta\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right) - \frac{m^2}{\sin^2\theta} = \frac{2M}{\hbar^2}[U(r) - E]r^2 - \frac{r}{R}\frac{\partial^2}{\partial r^2}(rR)$$

Again, since the LHS is independent of r and the RHS is independent of θ, then each side must be equal to a constant. Let us call the constant $-k$, then the LHS give us

$$\frac{1}{\Theta\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right) - \frac{m^2}{\sin^2\theta} = -k$$
$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right) + \left(k - \frac{m^2}{\sin^2\theta}\right)\Theta = 0$$

Finally, the RHS is

$$\begin{aligned}
\frac{2M}{\hbar^2}[U(r)-E]r^2 - \frac{r}{R}\frac{\partial^2}{\partial r^2}(rR) &= -k \\
\frac{r}{R}\frac{\partial^2}{\partial r^2}(rR) &= k + \frac{2M}{\hbar^2}[U(r)-E]r^2 \\
\frac{r}{R}\frac{\partial^2}{\partial r^2}(rR) &= \frac{2M}{\hbar^2}\left[\frac{k\hbar^2}{2M} + [U(r)-E]r^2\right] \\
\frac{\partial^2}{\partial r^2}(rR) &= \frac{2M}{\hbar^2}\left[\frac{k\hbar^2}{2Mr^2} + U(r) - E\right]rR
\end{aligned}$$

□

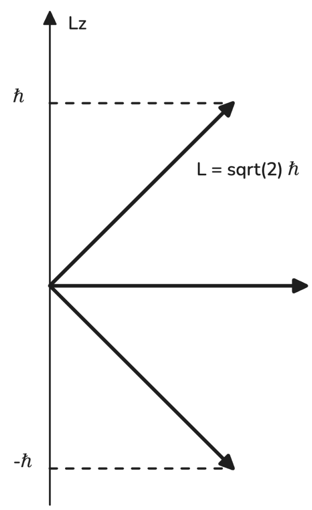
Solution. 8.23 The minimum possible angle between $\boldsymbol{L}$ and the z axis in the case $l = 2$ happens for $m = 2$ hence $L_z = 2\hbar$ so

$$\theta = \arccos\left(\frac{2\hbar}{\sqrt{2(2+1)}\hbar}\right) = 0.6154 \text{ rad} = 35.26^\circ$$

□

Solution. 8.24

(a) Angular momentum when $l = 1$.



(b) There are $(2l + 1) = 3$ possible orientations

(c) The minimum possible angle between $\boldsymbol{L}$ and the z axis in the case $l = 1$ happens for $m = 1$ hence $L_z = \hbar$ so

$$\theta = \arccos\left(\frac{1\hbar}{\sqrt{1(1+1)}\hbar}\right) = 0.7853 \text{ rad} = 45^\circ$$

□

Solution. 8.26 Given that the magnitude of $\boldsymbol{L}$ is $L = \sqrt{l(l+1)}\hbar$ then by definition L_z is given by $L_z = m\hbar$ where m can take values $-l, -l+1, ..., l-1, l$, so the biggest magnitude L_z can have is $L_z = l\hbar$.
We see that $l^2 \leq l^2 + l$ then $l \leq \sqrt{l(l+1)}$. Therefore the largest value of L_z is less than or equal to L. $\square$

Solution. 8.27 The ratios L/L_{bohr} are

$$
\begin{array}{ll}
l = 1 & \dfrac{L}{L_{bohr}} = \dfrac{\sqrt{1(1+1)}\hbar}{1\hbar} = 1.414 \\
l = 2 & \dfrac{L}{L_{bohr}} = \dfrac{\sqrt{2(2+1)}\hbar}{2\hbar} = 1.224 \\
l = 3 & \dfrac{L}{L_{bohr}} = \dfrac{\sqrt{3(3+1)}\hbar}{3\hbar} = 1.154 \\
l = 4 & \dfrac{L}{L_{bohr}} = \dfrac{\sqrt{4(4+1)}\hbar}{4\hbar} = 1.118 \\
l = 10 & \dfrac{L}{L_{bohr}} = \dfrac{\sqrt{10(10+1)}\hbar}{10\hbar} = 1.048 \\
l = 100 & \dfrac{L}{L_{bohr}} = \dfrac{\sqrt{100(100+1)}\hbar}{100\hbar} = 1.004
\end{array}
$$

We see that as l grows then L/L_{bohr} tends to 1 which implies that for big l then L_{bohr} is quite close to the correct L.

$\square$

Solution. 8.28 The allowed magnitudes of the angular momentum are

$$L = \sqrt{l(l+1)}\hbar = \sqrt{l^2\left(1 + \frac{1}{l}\right)}\hbar = l\sqrt{1 + \frac{1}{l}}\hbar$$

If l is large then we can approximate the square root using the binomial expansion as follows

$$L \approx l\left(1 + \frac{1}{2l}\right)\hbar = \left(l + \frac{1}{2}\right)\hbar$$

$\square$

Solution. 8.29 The equation for θ when $m = l = 0$ becomes

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) = 0$$

(a) Let $\Theta = C$ where C is a constant then $d\Theta/d\theta = d(C)/d\theta = 0$ so the equation is satisfied.

(b) Let

$$\Theta = \ln\left(\frac{1+\cos\theta}{1-\cos\theta}\right)$$

Then

$$\frac{\mathrm{d}\Theta}{\mathrm{d}\theta} = -\frac{2}{\sin\theta}$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) = \frac{\mathrm{d}}{\mathrm{d}\theta}(-2) = 0$$

So the function Θ is a solution of the θ equation.

If we let $\theta = 0$ we see that

$$\Theta = \ln\left(\frac{1+1}{1-1}\right) \to \infty$$

and if we let $\theta = \pi$ we get that

$$\Theta = \ln\left(\frac{1+(-1)}{1-(-1)}\right) \to -\infty$$

Since $\lim_{x\to\infty}\ln(x) = \infty$ and $\lim_{x\to 0}\ln(x) = -\infty$.

(c) Then the general solution for the θ equation must be a linear combination of both solutions

$$\Theta = aC + b\ln\left(\frac{1+\cos\theta}{1-\cos\theta}\right)$$

where a, b are constants. Again for $\theta = 0$ or π we get that $\Theta \to \infty$ or $\Theta \to -\infty$ respectively, so the second term cannot be part of the general solution since it's not well defined for every θ. Therefore only the solution $\Theta = C$ is an acceptable solution.

□

Solution. 8.30 The θ equation for the case $l = m = 1$ is

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) + \left(2 - \frac{1}{\sin^2\theta}\right)\Theta = 0$$

Let $\Theta = \sin\theta$ then we see that

$$\begin{aligned}
\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) + \left(2 - \frac{1}{\sin^2\theta}\right)\Theta &= \\
&= \frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}(\sin\theta\cos\theta) + \left(2 - \frac{1}{\sin^2\theta}\right)\sin\theta \\
&= \frac{1}{\sin\theta}(\cos^2\theta - \sin^2\theta) + 2\sin\theta - \frac{1}{\sin\theta} \\
&= \frac{\cos^2\theta}{\sin\theta} + \sin\theta - \frac{1}{\sin\theta} \\
&= \frac{\cos^2\theta + \sin^2\theta - 1}{\sin\theta} \\
&= \frac{1-1}{\sin\theta} \\
&= 0
\end{aligned}$$

So $\Theta = \sin\theta$ is a solution of the θ equation.
The complete wave functions for the cases $l = 1$, $m = 1$ and $l = 1$, $m = -1$ are

$$\psi(r,\theta,\phi) = R(r)\sin\theta e^{i\phi} \qquad \psi(r,\theta,\phi) = R(r)\sin\theta e^{-i\phi}$$

Assuming r fixed then ψ only depends on θ and ϕ but computing $|\psi|^2$ we see that

$$|\psi|^2 \propto \sin^2\theta|e^{i\phi}|^2 = \sin^2\theta$$

Since $|e^{i\phi}|^2 = 1$, then $|\psi|^2$ is maximum for $\theta = \pi/2$.
In the case of $\psi(r,\theta,\phi) = R(r)\sin\theta e^{-i\phi}$ since $|e^{-i\phi}|^2 = 1$ we get that $|\psi|^2$ is maximum for $\theta = \pi/2$ as well.

□

Solution. 8.31 Let $\Theta_{l,m}(\theta)$ be a solution to the θ equation

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right)+\left(l(l+1)-\frac{m^2}{\sin^2\theta}\right)\Theta=0$$

The θ equation for $l,-m$ is the same since it depends on $(-m)^2=m^2$, then the θ equation for $l,-m$ must also be solved by $\Theta_{l,m}(\theta)$ but the θ equation has at most one independent acceptable solution for any given values of l and m, so must be that the solution $\Theta_{l,-m}(\theta)$ is at most proportional to $\Theta_{l,m}(\theta)$ i.e. $\Theta_{l,-m}(\theta)\propto\Theta_{l,m}(\theta)$. $\square$

Solution. 8.35
The degeneracy of the nth level is given by $1+3+5+...+(2n-1)$, we can write this sum as $\sum_{k=1}^{n} 2k-1$ then

$$\sum_{k=1}^{n} 2k-1 = 2\sum_{k=1}^{n} k - \sum_{k=1}^{n} 1 = 2 \cdot \frac{n(n+1)}{2} - n = n^2$$

□

Solution. 8.36
If certain hydrogen atom has a definite value of l then

(a) The angular momentum has a magnitude $L = \sqrt{l(l+1)}\hbar$ and the z component of L is $L_z = m\hbar$ where m can take the following values $m = l, l-1, ..., 0, ... -l+1, -l$.

(b) The allowed energies consistent with this information are

$$E = -\frac{E_R}{n^2}$$

where $n > l$.

□

Solution. 8.37
Let us compute the mean value (or expectation value) of $1/r$ for the $1s$ state as follows

$$\begin{aligned}
\langle 1/r \rangle &= \int_0^\infty \frac{1}{r} \cdot P(r)\ dr \\
&= \int_0^\infty \frac{1}{r} \cdot 4\pi A^2 r^2 e^{-2r/a_B}\ dr \\
&= 4\pi A^2 \int_0^\infty r e^{-2r/a_B}\ dr \\
&= \pi A^2 a_B^2
\end{aligned}$$

Where we used that $\int_0^\infty r e^{-2r/a_B}\ dr = a_B^2/4$. Also, by the normalization condition we know that $A^2 = 1/(\pi a_B^3)$, then

$$\langle 1/r \rangle = \frac{1}{a_B}$$

□

Solution. 8.38

(a) If for some hydrogen atom $n = 5$ then l can only take the values $0, 1, 2, 3$ or 4. But if $l = 1$ for example, then m can only take the values $-1, 0$ or 1, so the only states consistent with $m = 2$ are the states where l is $l \geq 2$ i.e. 2, 3 or 4 since only in those states m can have the value 2. Therefore there are 3 states consistent with $n = 5$ and $m = 2$.

(b) In the general case, if we have an hydrogen atom with number n then l can only take the numbers

$$0, 1, ..., n-1$$

So if we are given a number $m < n$ then the only states consistent with n and m are those where $l \geq m$ so from the possible values of l we need to remove those less than m.

The list of possible values of l is of size $n-1$ and removing the $m-1$ terms which are less than m we get that

$$n - 1 - (m-1) = n - m$$

Therefore the states consistent with n and m are $n - m$.

□

Solution. 8.39

(a) Equation (8.79) recalling that $a_B = \hbar^2/(m_e k e^2)$ and that $E_R = ke^2/2a_B$ becomes

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}r^2}(rR) &= \frac{2m_e}{\hbar^2}\left[\frac{E_R}{n^2} - \frac{ke^2}{r}\right](rR) \\ &= \frac{2m_e}{\hbar^2}\left[\frac{ke^2}{2a_B n^2} - \frac{ke^2}{r}\right](rR) \\ &= \left[\frac{m_e k e^2}{a_B \hbar^2 n^2} - \frac{2m_e k e^2}{\hbar^2 r}\right](rR) \\ &= \left[\frac{1}{a_B^2 n^2} - \frac{2}{a_B r}\right](rR)\end{aligned}$$

(b) If $n = 1$ the equation becomes

$$\frac{\mathrm{d}^2}{\mathrm{d}r^2}(rR) = \left[\frac{1}{a_B^2} - \frac{2}{a_B r}\right](rR)$$

Let us check if $R_{1s} = e^{-r/a_B}$ is a solution. We see that

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}r^2}(rR_{1s}) &= \frac{\mathrm{d}^2}{\mathrm{d}r^2}\left(re^{-r/a_B}\right) \\ &= \frac{\mathrm{d}}{\mathrm{d}r}\left(e^{-r/a_B} - \frac{r}{a_B}e^{-r/a_B}\right) \\ &= -\frac{e^{-r/a_B}}{a_B} - \frac{1}{a_B}\left(e^{-r/a_B} - \frac{r}{a_B}e^{-r/a_B}\right) \\ &= -\frac{2e^{-r/a_B}}{a_B} + \frac{r}{a_B^2}e^{-r/a_B} \\ &= \left[\frac{1}{a_B^2} - \frac{2}{a_B r}\right](re^{-e/a_B}) \\ &= \left[\frac{1}{a_B^2} - \frac{2}{a_B r}\right](rR_{1s})\end{aligned}$$

So we see that indeed $R_{1s} = e^{-r/a_B}$ is a solution.

□

Solution. 8.40

(a) Equation (8.72) when $l = n-1$ becomes

$$\frac{\mathrm{d}^2}{\mathrm{d}r^2}(rR) = \frac{2m_e}{\hbar^2}\left[-\frac{ke^2}{r} + \frac{(n-1)n\hbar^2}{2m_e r^2} - E\right](rR)$$

(b) Let $R(r) = Ar^{n-1}e^{-r/na_B}$ then we see that

$$\begin{aligned}
\frac{\mathrm{d}^2}{\mathrm{d}r^2}(rR) &= \frac{\mathrm{d}^2}{\mathrm{d}r^2}Ar^n e^{-r/na_B} \\
&= A\frac{\mathrm{d}}{\mathrm{d}r}\left[nr^{n-1}e^{-r/na_B} - \frac{r^n}{na_B}e^{-r/na_B}\right] \\
&= A\left[n(n-1)r^{n-2}e^{-r/na_B} - \frac{r^{n-1}}{a_B}e^{-r/na_B} - \frac{r^{n-1}}{a_B}e^{-r/na_B} + \frac{r^n}{n^2a_B^2}e^{-r/na_B}\right] \\
&= A\left[\frac{n(n-1)}{r^2} - \frac{2}{a_B r} + \frac{1}{n^2a_B^2}\right]r^n e^{-r/na_B} \\
&= A\left[\frac{n(n-1)}{r^2} - \frac{2ke^2m}{\hbar^2 r} + \frac{k^2e^4m^2}{n^2\hbar^4}\right]r^n e^{-r/na_B} \\
&= \frac{2m}{\hbar^2}\left[\frac{n(n-1)\hbar^2}{2mr^2} - \frac{ke^2}{r} + \frac{k^2e^4m}{2n^2\hbar^2}\right]Ar^n e^{-r/na_B} \\
&= \frac{2m}{\hbar^2}\left[\frac{n(n-1)\hbar^2}{2mr^2} - \frac{ke^2}{r} + \frac{E_R}{n^2}\right]rR
\end{aligned}$$

So if $E = -E_R/n^2$ then the $Ar^{n-1}e^{-r/na_B}$ is a solution.

$\square$

Solution. 8.41 Let $u = r^2$ and $dv = e^{-2r/a_B}\ dr$ then integrating by parts the equation (8.87) gives us

$$\begin{aligned}
4\pi A^2 \int_0^\infty r^2 e^{-2r/a_B}\ dr &= 4\pi A^2 \left[\frac{-a_B r^2}{2} e^{-2r/a_B} \Big|_0^\infty + a_B \int_0^\infty r e^{-2r/a_B}\ dr \right] \\
&= 4\pi A^2 a_B \int_0^\infty r e^{-2r/a_B}\ dr
\end{aligned}$$

Now, applying integration by parts again taking $u = r$ and $dv = e^{-2r/a_B}\ dr$ we get that

$$\begin{aligned}
4\pi A^2 a_B \int_0^\infty r e^{-2r/a_B}\ dr &= 4\pi A^2 a_B \left[\frac{-a_B r}{2} e^{-2r/a_B} \Big|_0^\infty + \frac{a_B}{2} \int_0^\infty e^{-2r/a_B}\ dr \right] \\
&= 2\pi A^2 a_B^2 \int_0^\infty e^{-2r/a_B}\ dr \\
&= -\pi A^2 a_B^3 e^{-2r/a_B} \Big|_0^\infty \\
&= \pi A^2 a_B^3
\end{aligned}$$

Therefore, since $4\pi A^2 \int_0^\infty r^2 e^{-2r/a_B}\ dr = \pi A^2 a_B^3 = 1$ we get that

$$A = \frac{1}{\sqrt{\pi a_B^3}}$$

$\square$

Solution. 8.42 We know that for the $1s$ state of the hydrogen

$$P_{1s}(r) = 4\pi A^2 r^2 e^{-2r/a_B}$$

Then the average value of $\langle r \rangle$ using integration by parts is given by

$$\begin{aligned}
\langle r \rangle &= \int_0^\infty r P_{1s}(r)\ dr \\
&= 4\pi A^2 \int_0^\infty r^3 e^{-2r/a_B}\ dr \\
&= 4\pi A^2 \left[-\frac{r^3 a_B}{2} e^{-2r/a_B} \Big|_0^\infty + \frac{3a_B}{2} \int_0^\infty r^2 e^{-2r/a_B}\ dr \right] \\
&= 6\pi A^2 a_B \left[-\frac{a_B r^2}{2} e^{-2r/a_B} \Big|_0^\infty + a_B \int_0^\infty r e^{-2r/a_B}\ dr \right] \\
&= 6\pi A^2 a_B^2 \left[-\frac{a_B r}{2} e^{-2r/a_B} \Big|_0^\infty + \frac{a_B}{2} \int_0^\infty e^{-2r/a_B}\ dr \right] \\
&= 6\pi A^2 a_B^2 \left[-\frac{a_B^2}{4} e^{-2r/a_B} \right]_0^\infty \\
&= 3\pi \frac{1}{\pi a_B^3} \frac{a_B^4}{2} \\
&= \frac{3a_B}{2}
\end{aligned}$$

Comparing to Fig 8.18 the most probable radii is a_B the reason is that $\langle r \rangle$ is the value we would get if we take a lot of measurements which is not the same as the most probable radii. $\square$

Solution. 8.43 The probability that a $1s$ electron in hydrogen would be found outside the Bohr radius is

$$\begin{aligned}
\int_{a_B}^{\infty} P(r)\ dr &= \frac{4}{a_B^3}\int_{a_B}^{\infty} r^2 e^{-2r/a_B}\ dr \\
&= \frac{4}{a_B^3}\left[-\frac{r^2 a_B}{2}e^{-2r/a_B}\Big|_{a_B}^{\infty} + a_B\int_{a_B}^{\infty} re^{-2r/a_B}\ dr\right] \\
&= \frac{4}{a_B^3}\left[\frac{a_B^3}{2}e^{-2} + a_B\int_{a_B}^{\infty} re^{-2r/a_B}\ dr\right] \\
&= \frac{4}{a_B^3}\left[\frac{a_B^3}{2}e^{-2} + a_B\left[-\frac{a_B r}{2}e^{-2r/a_B}\Big|_{a_B}^{\infty} + \frac{a_B}{2}\int_{a_B}^{\infty} e^{-2r/a_B}\ dr\right]\right] \\
&= \frac{4}{a_B^3}\left[a_B^3 e^{-2} + \frac{a_B^2}{2}\int_{a_B}^{\infty} e^{-2r/a_B}\ dr\right] \\
&= \frac{4}{a_B^3}\left[a_B^3 e^{-2} - \frac{a_B^3}{4}e^{-2r/a_B}\Big|_{a_B}^{\infty}\right] \\
&= \frac{4}{a_B^3}\left[\frac{5a_B^3}{4}e^{-2}\right] \\
&= 5e^{-2} \approx 0.67
\end{aligned}$$

$\square$

Solution. 8.44

(a) The radial equation (8.107) for the case $n = 2$ and $l = 0$ is

$$\frac{\mathrm{d}^2}{\mathrm{d}r^2}(rR) = \left(\frac{1}{4a_B^2} - \frac{2}{a_B r}\right)(rR)$$

Let us verify that

$$R_{2s} = A\left(2 - \frac{r}{a_B}\right)e^{-r/2a_B}$$

is a solution to the equation as follows

$$\begin{aligned}
\frac{\mathrm{d}^2}{\mathrm{d}r^2}(rR_{2s}) &= A\frac{\mathrm{d}^2}{\mathrm{d}r^2}\left(2r - \frac{r^2}{a_B}\right)e^{-r/2a_B} \\
&= A\frac{\mathrm{d}}{\mathrm{d}r}\left[\left(2 - \frac{2r}{a_B}\right)e^{-r/2a_B} - \frac{1}{2a_B}\left(2r - \frac{r^2}{a_B}\right)e^{-r/2a_B}\right] \\
&= A\left[-\frac{2}{a_B} - \frac{1}{2a_B}\left(2 - \frac{2r}{a_B}\right) - \frac{1}{2a_B}\left(2 - \frac{2r}{a_B}\right) + \frac{1}{4a_B^2}\left(2r - \frac{r^2}{a_B}\right)\right]e^{-r/2a_B} \\
&= A\left[-\frac{2}{a_B} - \frac{1}{a_B} + \frac{r}{a_B^2} - \frac{1}{a_B} + \frac{r}{a_B^2} + \frac{r}{2a_B^2} - \frac{r^2}{4a_B^3}\right]e^{-r/2a_B} \\
&= A\left[-\frac{4}{a_B r} + \frac{5}{2a_B^2} - \frac{r}{4a_B^3}\right]re^{-r/2a_B}
\end{aligned}$$

Let us note that

$$\begin{aligned}
\left(\frac{1}{4a_B^2} - \frac{2}{a_B r}\right)\left(2 - \frac{r}{a_B}\right) &= \frac{1}{2a_B^2} - \frac{r}{4a_B^3} - \frac{4}{a_B r} + \frac{2}{a_B^2} \\
&= -\frac{4}{a_B r} + \frac{5}{2a_B^2} - \frac{r}{4a_B^3}
\end{aligned}$$

So

$$\frac{\mathrm{d}^2}{\mathrm{d}r^2}(rR_{2s}) = A\left(\frac{1}{4a_B^2} - \frac{2}{a_B r}\right)\left(2 - \frac{r}{a_B}\right)re^{-r/2a_B} = \left(\frac{1}{4a_B^2} - \frac{2}{a_B r}\right)(rR_{2s})$$

Therefore R_{2s} is a solution to the radial equation.

(b) The normalization condition in this case gives us

$$\begin{aligned}
4\pi A^2\int_0^\infty r^2\left(2 - \frac{r}{a_B}\right)^2 e^{-r/a_B}\ dr &= 1 \\
4\pi A^2\int_0^\infty 4r^2e^{-r/a_B} - \frac{4r^3}{a_B}e^{-r/a_B} + \frac{r^4}{a_B^2}e^{-r/a_B}\ dr &= 1 \\
4\pi A^2\left[8a_B^3 - 24a_B^3 + 24a_B^3\right] &= 1 \\
32\pi A^2 a_B^3 &= 1 \\
A &= \frac{1}{\sqrt{32xgs\pi a_B^3}}
\end{aligned}$$

$\square$

Solution. 8.45
Let $n=2$ and $l=1$ then the radial equation becomes

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}r^2}(rR) &= \frac{2m_e}{\hbar^2}\left[-\frac{ke^2}{r}+\frac{2\hbar^2}{2m_er^2}+\frac{E_R}{4}\right](rR)\\ &= \left[-\frac{2}{a_Br}+\frac{2}{r^2}+\frac{1}{4a_B^2}\right](rR)\end{aligned}$$

Let us verify now that $R_{2p}=Are^{-r/2a_B}$ is a solution to the radial equation

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}r^2}(rR_{2p}) &= A\frac{\mathrm{d}^2}{\mathrm{d}r^2}r^2e^{-r/2a_B}\\ &= A\frac{\mathrm{d}}{\mathrm{d}r}\left[2re^{-r/2a_B}-\frac{r^2}{2a_B}e^{-r/2a_B}\right]\\ &= A\left[2e^{-r/2a_B}-\frac{r}{a_B}e^{-r/2a_B}-\frac{r}{a_B}e^{-r/2a_B}+\frac{r^2}{4a_B^2}e^{-r/2a_B}\right]\\ &= A\left[\frac{2}{r^2}-\frac{2}{a_Br}+\frac{1}{4a_B^2}\right]r^2e^{-r/2a_B}\\ &= \left[\frac{2}{r^2}-\frac{2}{a_Br}+\frac{1}{4a_B^2}\right](rR_{2p})\end{aligned}$$

Therefore R_{2p} is a solution to the radial equation.
Let $P_{2p}=4\pi r^2|R_{2p}|^2$ then the normalization condition gives us

$$4\pi A^2\int_0^\infty r^4e^{-r/a_B}\ dr=1$$

Let $u=r/a_B$ then $du=dr/a_B$ so

$$\begin{aligned}4\pi A^2a_B^5\int_0^\infty u^4e^{-u}\ du&=1\\ 4\pi A^2a_B^5\ 24&=1\\ A&=\frac{1}{4\sqrt{6\pi a_B^5}}\end{aligned}$$

Where we used the known solution $\int_0^\infty u^4e^{-u}du=24$. □

Solution. 8.46

(a) We know that R_{2p} is given by

$$R_{2p} = \frac{1}{4\sqrt{6\pi a_B^5}} r e^{-r/2a_B}$$

Then $P(r)$ is given by

$$P(r) = 4\pi r^2 R_{2p}^2 = \frac{1}{24 a_B^5} r^4 e^{-r/a_B}$$

Then the expectation value of the radius is

$$\begin{aligned}
\langle r \rangle &= \int_0^\infty r P(r)\ dr \\
&= \frac{1}{24 a_B^5} \int_0^\infty r^5 e^{-r/a_B}\ dr
\end{aligned}$$

Let $u = r/a_B$ then $du = dr/a_B$ so

$$\langle r \rangle = \frac{a_B}{24} \int_0^\infty u^5 e^{-u}\ du = \frac{a_B}{24} \cdot 5! = 5a_B$$

(b) The average potential energy is given by

$$\begin{aligned}
\langle U \rangle &= \int_0^\infty U(r) P(r)\ dr \\
&= -\frac{ke^2}{24 a_B^5} \int_0^\infty r^3 e^{-r/a_B}\ dr
\end{aligned}$$

Again, let $u = r/a_B$ then $du = dr/a_B$ so

$$\langle U \rangle = -\frac{ke^2}{24 a_B^5} \int_0^\infty a_B^4 u^3 e^{-u}\ du = -\frac{ke^2}{24 a_B} \cdot 3! = -\frac{ke^2}{4a_B}$$

(c) According to the Bohr model the potential energy should be

$$U = -\frac{ke^2}{n a_B} = -\frac{ke^2}{2a_B}$$

So the average potential energy in this case is half what the Bohr model says.

□

Solution. 8.47

(a) The θ equation for the $2p$ state with $m = \pm 1$ is

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) + \left(2 - \frac{1}{\sin^2\theta}\right)\Theta = 0$$

Let $\Theta = \sin\theta$ then we see that

$$\begin{aligned}
&\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) + \left(2 - \frac{1}{\sin^2\theta}\right)\Theta = \\
&= \frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}(\sin\theta\cos\theta) + 2\sin\theta - \frac{1}{\sin\theta} \\
&= \frac{1}{\sin\theta}(\cos^2\theta - \sin^2\theta) + 2\sin\theta - \frac{1}{\sin\theta} \\
&= \frac{\cos^2\theta}{\sin\theta} + \sin\theta - \frac{1}{\sin\theta} \\
&= \frac{1}{\sin\theta}(\cos^2\theta - 1) + \sin\theta \\
&= -\frac{\sin^2\theta}{\sin\theta} + \sin\theta \\
&= -\sin\theta + \sin\theta \\
&= 0
\end{aligned}$$

Therefore $\Theta = \sin\theta$ is a solution to the θ equation.

(b) We know that the wave function $2p_x$ is given by

$$\psi_{2p_x} = Axe^{-r/2a_B}$$

Then, let us add the wave functions $\psi_{2,1,\pm1}$ as follows

$$\begin{aligned}
\psi_{2,1,1} + \psi_{2,1,-1} &= R_{2p}\sin\theta(e^{i\phi} + e^{-i\phi}) \\
&= R_{2p}\sin\theta(\cos\phi + i\sin\phi + \cos\phi - i\sin\phi) \\
&= 2R_{2p}\sin\theta\cos\phi
\end{aligned}$$

Now replacing $R_{2p} = Are^{-r/2a_B}$ we get that

$$\psi_{2,1,1} + \psi_{2,1,-1} = 2Are^{-r/2a_B}\sin\theta\cos\phi = 2Axe^{-r/2a_B}$$

Where we used taht $x = r\sin\theta\cos\phi$. We see that the sum of the two wave functions is ψ_{2p_x} times 2.

On the other hand, we know that $\psi_{2p_y} = Aye^{-r/2a_B}$. Taking the difference of the wave functions $\psi_{2,1,\pm1}$ we get that

$$\begin{aligned}
\psi_{2,1,1} - \psi_{2,1,-1} &= R_{2p}\sin\theta(e^{i\phi} - e^{-i\phi}) \\
&= R_{2p}\sin\theta(\cos\phi + i\sin\phi - \cos\phi + i\sin\phi) \\
&= 2iR_{2p}\sin\theta\sin\phi \\
&= 2iAre^{-r/2a_B}\sin\theta\sin\phi \\
&= 2iAye^{-r/2a_B}
\end{aligned}$$

Where we used that $y = r \sin\theta \sin\phi$. Therefore the difference of the two wave functions is ψ_{2p_y} times $2i$.

$\square$

Solution. 8.48
The probability density $P(r)$ will be maximum when $dP(r)/dr = 0$ and hence

$$
\begin{aligned}
\frac{\mathrm{d}P(r)}{\mathrm{d}r} &= \frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{4}{a_B^3}r^2e^{-2r/a_B}\right) \\
&= \frac{4}{a_B^3}\left(2re^{-2r/a_B} - \frac{2r^2}{a_B}e^{-2r/a_B}\right)
\end{aligned}
$$

Then, equating this equation to 0 we get that

$$
\begin{aligned}
\frac{4}{a_B^3}\left(2re^{-2r/a_B} - \frac{2r^2}{a_B}e^{-2r/a_B}\right) &= 0 \\
re^{-2r/a_B} - \frac{r^2}{a_B}e^{-2r/a_B} &= 0 \\
\frac{r^2}{a_B}e^{-2r/a_B} &= re^{-2r/a_B} \\
\frac{r^2}{a_B} &= r \\
r &= a_B
\end{aligned}
$$

Therefore $P(r)$ is maximum when $r = a_B$ i.e. the most probable radius is a_B. □

Solution. 8.49
The probability density for a hydrogen atom in a state where $l = n - 1$ is

$$P(r) = 4\pi r^2 \left[A r^{n-1} e^{-r/a_B} \right]^2 = 4\pi A^2 r^{2n} e^{-2r/a_B}$$

The most probable radius will happen where $dP(r)/dr = 0$ hence

$$\frac{\mathrm{d}P(r)}{\mathrm{d}r} = 4\pi A^2 \left[2nr^{2n-1} e^{-2r/a_B} - \frac{2r^{2n}}{a_B} e^{-2r/a_B} \right]$$

Then

$$\begin{aligned}
4\pi A^2 \left[2nr^{2n-1} e^{-2r/a_B} - \frac{2r^{2n}}{a_B} e^{-2r/a_B} \right] &= 0 \\
2nr^{2n-1} e^{-2r/a_B} - \frac{2r^{2n}}{a_B} e^{-2r/a_B} &= 0 \\
n\frac{1}{r} - \frac{1}{a_B} &= 0 \\
r &= na_B
\end{aligned}$$

Therefore, we see that this result agrees with the expected radius from the Bohr model. □

Solution. 8.50
The probability density for a hydrogen atom in a state $2s$ is

$$P_{2s}(r) = 4\pi r^2 \left[A\left(2 - \frac{r}{a_B}\right) e^{-r/2a_B} \right]^2$$

And for a hydrogen atom in state $2p$ is

$$P_{2p}(r) = 4\pi r^2 \left[A r e^{-r/2a_B} \right]^2$$

Given that $P(r)$ is maximum at the same r where $\sqrt{P(r)}$ is maximum then we can compute the most probable radius of $P_{2s}(r)$ as

$$\begin{aligned}
\frac{\mathrm{d}\sqrt{P_{2s}(r)}}{\mathrm{d}r} &= 0 \\
\frac{\mathrm{d}}{\mathrm{d}r}\left(\sqrt{4\pi} r A \left(2 - \frac{r}{a_B}\right) e^{-r/2a_B}\right) &= 0 \\
\sqrt{4\pi} A \left[\left(2 - \frac{2r}{a_B}\right) e^{-r/2a_B} - \left(2r - \frac{r^2}{a_B}\right) \frac{1}{2a_B} e^{-r/2a_B} \right] &= 0 \\
2 - \frac{2r}{a_B} - \frac{r}{a_B} + \frac{r^2}{2a_B^2} &= 0 \\
\frac{r^2}{2a_B} - 3r + 2a_B &= 0
\end{aligned}$$

The solutions to this quadratic equation are

$$r = \frac{3 \pm \sqrt{9 - 4 \cdot (1/2a_B) \cdot 2a_B}}{1/a_B} = (3 \pm \sqrt{5}) a_B$$

But from the plots of $P_{2s}(r)$ in problem 8.55 we see that the most probable radius is $r = (3 + \sqrt{5}) a_B \approx 5.2 a_B$.
For the case of $P_{2p}(r)$ we have that

$$\begin{aligned}
\frac{\mathrm{d}\sqrt{P_{2p}(r)}}{\mathrm{d}r} &= 0 \\
\frac{\mathrm{d}}{\mathrm{d}r}\left(\sqrt{4\pi} A r^2 e^{-r/2a_B}\right) &= 0 \\
\sqrt{4\pi} A \left(2r e^{-r/2a_B} - \frac{r^2}{2a_B} e^{-r/2a_B} \right) &= 0 \\
2r - \frac{r^2}{2a_B} &= 0 \\
r &= 4a_B
\end{aligned}$$

□

Solution. 8.51
The probability distribution for a $1s$ electron in the hydrogen-like ion Ni^{27+} is

$$P_{1s}(r) = \frac{4Z^3}{a_B^3} r^2 e^{-2Zr/a_B}$$

Where we replaced a_B with a_B/Z and Z in this case should be 28.
We know that the most probable radius will happen where $dP(r)/dr = 0$ hence

$$\begin{aligned}\frac{\mathrm{d}P(r)}{\mathrm{d}r} &= \frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{4Z^3}{a_B^3} r^2 e^{-2Zr/a_B}\right)\\ &= \frac{4Z^3}{a_B^3}\left(2re^{-2Zr/a_B} - \frac{2Zr^2}{a_B}e^{-2Zr/a_B}\right)\end{aligned}$$

Then, equating this to 0 we get that

$$\begin{aligned}\frac{4Z^3}{a_B^3}\left(2re^{-2Zr/a_B} - \frac{2Zr^2}{a_B}e^{-2Zr/a_B}\right) &= 0\\ re^{-2Zr/a_B} - \frac{Zr^2}{a_B}e^{-2Zr/a_B} &= 0\\ \frac{Zr^2}{a_B}e^{-2Zr/a_B} &= re^{-2Zr/a_B}\\ \frac{Zr^2}{a_B} &= r\\ r &= \frac{a_B}{Z}\end{aligned}$$

Therefore $P(r)$ is maximum when $r = a_B/Z$ i.e. the most probable radius is $a_B/28$.
The binding energy is given by

$$E = -Z^2 E_R = -784 E_R$$

□

Solution. 8.52

Assuming the electron has a wave function very like that for a single electron in orbit around the same nucleus, then the most probable radius for a $1s$ electron is $r = a_B/Z$ as we determined in Problem 8.51.

Also, the electron's approximate binding energy is $E = -Z^2 E_R$ as determined in Problem 8.51. $\square$

Solution. 8.53
The energy emitted when the electron drops from $n = 2$ to $n = 1$ is

$$E_\gamma = E_2 - E_1 = Z^2 E_R \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{3}{4} Z^2 E_R$$

Where $Z = 12$ for the case Mg^{11+}. Then using that $E_\gamma = hc/\lambda$ we get that the wavelength emitted is

$$\lambda = \frac{4hc}{3Z^2 E_R} = \frac{4 \cdot 1240}{3 \cdot (12^2) \cdot 13.6} = 0.844\ nm$$

This wavelength falls within the X-ray wavelength. $\square$

Solution. 8.54
The energy emitted when the electron drops from state n to state m is

$$E_\gamma = E_n - E_m = Z^2 E_R \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

We want to determine n and m such that $E_\gamma = 756\ eV$ then we see that

$$\frac{1}{m^2} - \frac{1}{n^2} = \frac{E_\gamma}{Z^2 E_R} = \frac{756}{(20^2)13.6} = 0.139$$

Also, we see that if $m = 1$ and $n = 2$ then

$$\frac{1}{1^2} - \frac{1}{2^2} = 0.75 > 0.139$$

and if $n = 3, 4, ...$ then

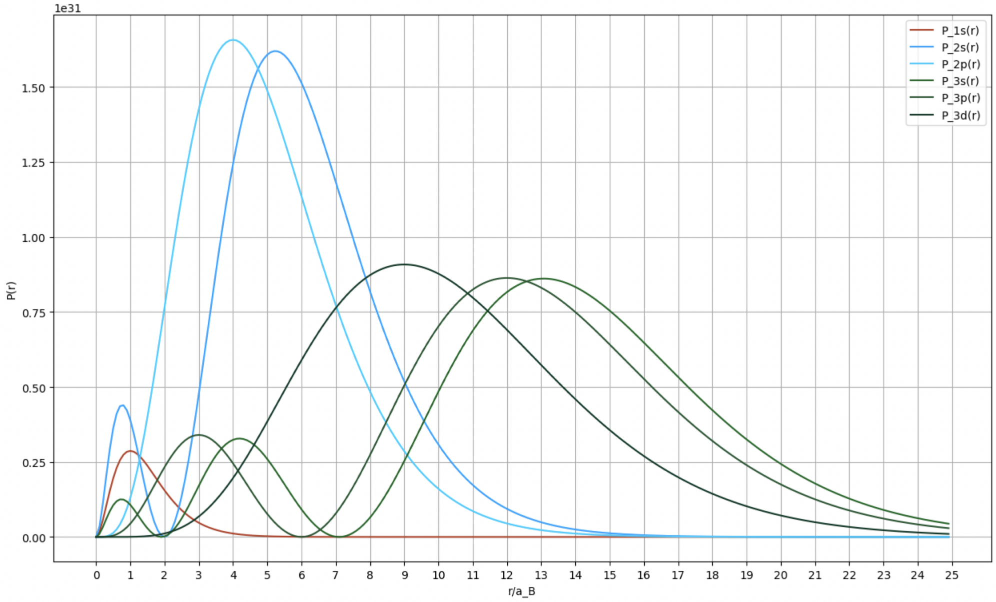
$$\frac{1}{m^2} - \frac{1}{n^2} > 0.75 > 0.139$$

So the electron cannot drop to state $m = 1$. If we try $m = 2$ with different n-s we see that for $n = 3$ we get that

$$\frac{1}{2^2} - \frac{1}{3^2} = 0.139$$

Therefore the drop must be from level $n = 3$ to level $m = 2$. $\square$

Solution. 8.55 Below we show the radial probability distributions different hydrogen states

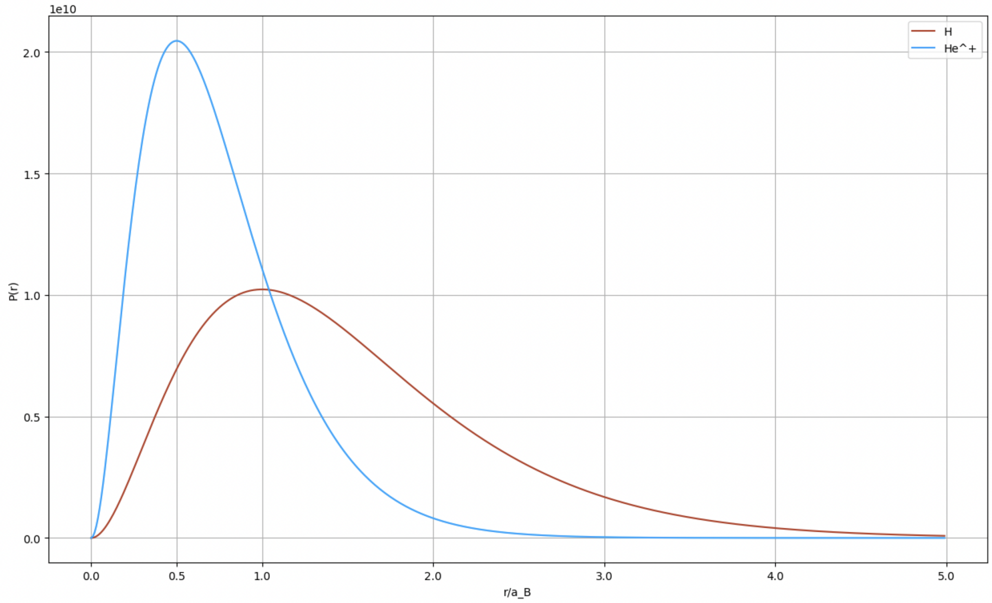


□

Solution. 8.56
Let us define $\rho = r/a_B$ then the radial probability distribution equation becomes

$$P(r) = \frac{4K^3 r^2}{a_B^3} e^{-2Kr/a_B} = \frac{4K^3}{a_B} \rho^2 e^{-2K\rho}$$

Using this formula we show below the radial probability distribution for Hydrogen and He^+



$\square$

Solution. 8.57

(a) Let us write equation (8.65) when $m = 0$ and $l = 2$

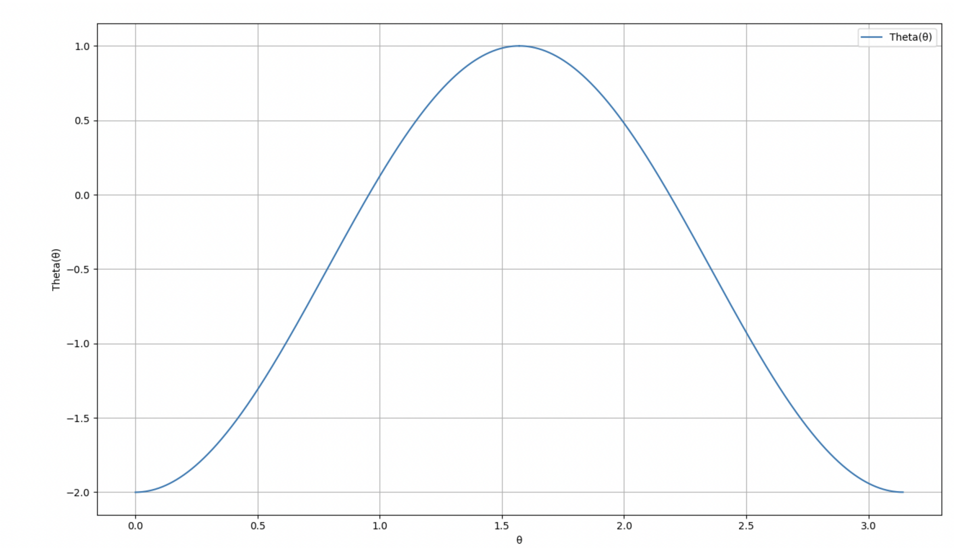
$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) + 6\Theta = 0$$

$$\frac{\mathrm{d}^2\Theta}{\mathrm{d}\theta^2} + \cot\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta} + 6\Theta = 0$$

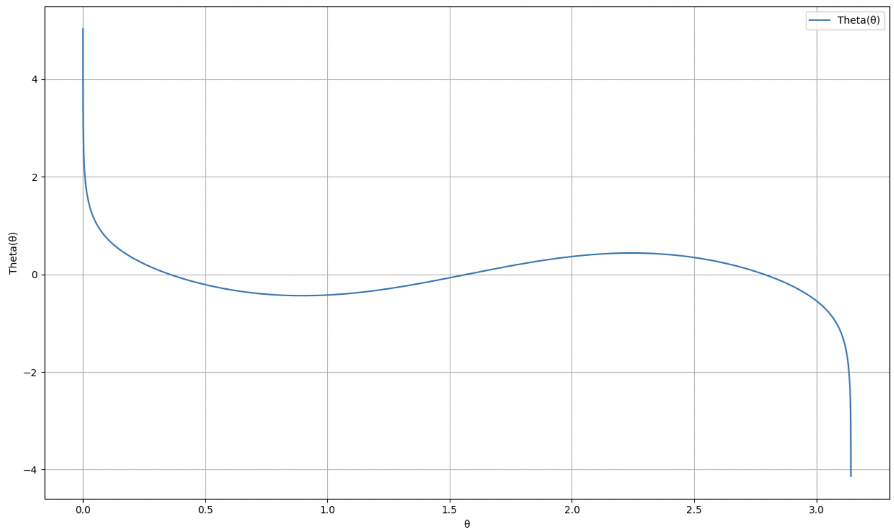
We can solve the equation numerically by applying the Runge-Kutta method, doing the following replacement. Let $u = \frac{\mathrm{d}\Theta}{\mathrm{d}\theta}$ then we get the following system of equations

$$\frac{\mathrm{d}\Theta}{\mathrm{d}\theta} = u \quad \text{and} \quad \frac{\mathrm{d}u}{\mathrm{d}\theta} = -u\cot\theta - 6\Theta$$

Then for the case $\Theta(\pi/2) = 1$ and $\Theta'(\pi/2) = 0$ we get that



For the case $\Theta(\pi/2) = 0$ and $\Theta'(\pi/2) = 1$ we get that

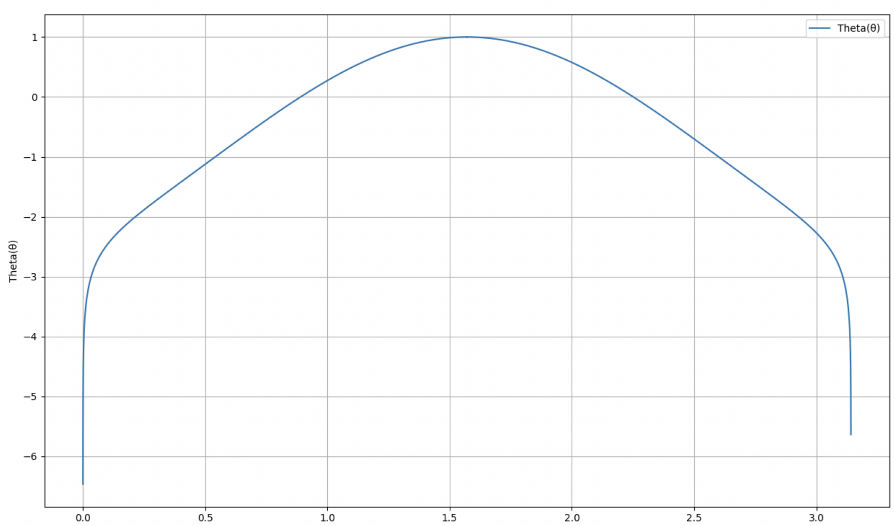


We see that this solution seems unacceptable because Θ at π and 0 seems to go to infinity.

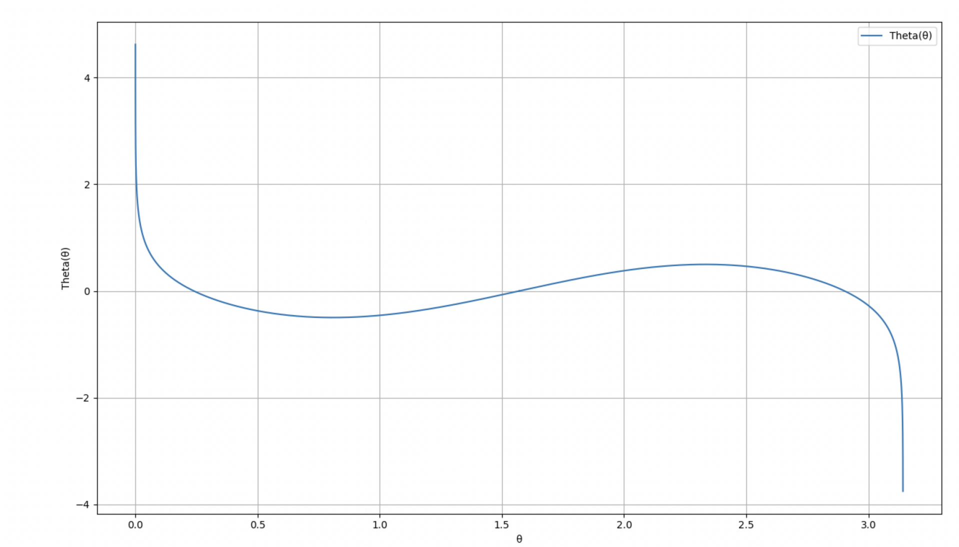
(b) Let us write equation (8.65) when $m = 0$ and $l = 1.75$

$$\frac{\mathrm{d}^2\Theta}{\mathrm{d}\theta^2} + \cot\theta \frac{\mathrm{d}\Theta}{\mathrm{d}\theta} + 4.812 \cdot \Theta = 0$$

Then solving by Runge-Kutta for the initial conditions $\Theta(\pi/2) = 1$ and $\Theta'(\pi/2) = 0$ we get that



And for the initial conditions $\Theta(\pi/2) = 0$ and $\Theta'(\pi/2) = 1$ we get that



In both cases the solution seems to diverge as $\theta \to \pi$ and as $\theta \to 0$.

$\square$